

===== :

Pandas Intro Basic Coding Day 1 Task

===== :



What is Pandas?

Pandas is a Python library.

Pandas is used to analyze data.

What Do Pandas?

Pandas allows us to analyze big data and make conclusions based on statistical theories.

Pandas can clean messy data sets, and make them readable and relevant.

Relevant data is very important in data science.

Why Use?

Read Data, CSV, EXCEL, SQL

Clean missing data

Filter and Sorting data

Analys Large dataset

Creat report easily.

Import Pandas:

Once Pandas is installed, import it in your applications by adding the import keyword:

Add your dataset to analyse

```
In [3]: # Dataset name = Country Economy analysis
```

1) Import Pandas

```
In [18]: import pandas as pd
```

2) Create Series

```
In [62]: s = pd.Series([10, 20, 30, 40, 50]) # series is a one Diamentional structure.  
print(s)
```

```
0    10  
1    20  
2    30  
3    40  
4    50  
dtype: int64
```

3) Create Dataframe :

```
In [43]: data = {  
    "Name": ["Azim", "SRK", "Priya"],  
    "Age": [25, 30, 24],  
    "City": ["Hydrabad", "Mumbai", "Delhi"]  
}  
  
df1 = pd.DataFrame(data) # A Dataframe is a table with rows and columns.  
print(df1)
```

```
   Name  Age  City  
0  Azim   25  Hydrabad  
1   SRK   30   Mumbai  
2  Priya   24    Delhi
```

4) Read Excel file

```
In [44]: df = pd.read_excel("E:/FSDS/data.xlsx")
```

```
In [45]: df
```

Out[45]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

5) Head()-- Show First Rows

In [63]: `df.head()`

Out[63]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

6) id() -- It show adress of this dataframe.

In [48]: `id(df)`

Out[48]: 2275457854608

7) version --- it show version of DF.

```
In [49]: pd.__version__
```

```
Out[49]: '2.3.3'
```

8) len() --- It show total length of DF.

```
In [71]: len(df)
```

```
Out[71]: 195
```

9) Shape() -- IT Check shape of DF.

```
In [51]: df.shape
```

```
Out[51]: (195, 5)
```

```
In [52]: df.head(2)
```

```
Out[52]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

```
In [53]: df.head(5)
```

```
Out[53]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

10) tail() -- it show last Rows

```
In [64]: df.tail()
```

Out[64]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [72]: `df.tail(4)`

Out[72]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [73]: `df.tail(2)`

Out[73]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

11) Info() --- It show lot of information display on screen like Data types,null value,columns details,memory usages.

In [228... `df`

Out[228...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [75]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName     195 non-null   object
1   CountryCode     195 non-null   object
2   BirthRate       195 non-null   float64
3   InternetUsers   195 non-null   float64
4   IncomeGroup     195 non-null   object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

12) Column --- It displays no. of column information

In [69]: `df.columns`

```
Out[69]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
               'IncomeGroup'],
              dtype='object')
```

In [70]: `len(df.columns)` # it displays total length of no. of column.

Out[70]: 5

13) dtypes --- Shows data type of every column

In [77]: `df.dtypes`

Out[77]: CountryName object
CountryCode object
BirthRate float64
InternetUsers float64
IncomeGroup object
dtype: object

14) Columns

In [78]: `df.columns`

Out[78]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
 'IncomeGroup'],
 dtype='object')

15) Select Single Column

In [229... `df`

Out[229]...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [79]: `df['CountryName']`

```
Out[79]: 0          Aruba
1    Afghanistan
2          Angola
3          Albania
4  United Arab Emirates
...
190    Yemen, Rep.
191    South Africa
192  Congo, Dem. Rep.
193          Zambia
194          Zimbabwe
Name: CountryName, Length: 195, dtype: object
```

In [80]: `df['CountryCode']`

```
Out[80]: 0      ABW
1      AFG
2      AGO
3      ALB
4      ARE
...
190    YEM
191    ZAF
192    COD
193    ZMB
194    ZWE
Name: CountryCode, Length: 195, dtype: object
```


16) Select Multiple columns

```
In [83]: df[['BirthRate', 'InternetUsers']]
```

```
Out[83]:
```

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985	19.1
3	12.877	57.2
4	11.044	88.0
...	...	...
190	32.947	20.0
191	20.850	46.5
192	42.394	2.2
193	40.471	15.4
194	35.715	18.5

195 rows × 2 columns

17) Select Specfic Rows and Columns

Selecting Rows by using loc

```
In [105... df.loc[0:5,['BirthRate','InternetUsers']] # it uses labels,name based
```

```
Out[105...
```

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985	19.1
3	12.877	57.2
4	11.044	88.0
5	17.716	59.9

18) Select Rows using iloc[]

```
In [106... df.iloc[0] # It uses index position,integer based.
```

```
Out[106... CountryName      Aruba
CountryCode      ABW
BirthRate      10.244
InternetUsers      78.9
IncomeGroup      High income
Name: 0, dtype: object
```

```
In [112... df.iloc[0:3,0:5]
```

```
Out[112... CountryName CountryCode BirthRate InternetUsers IncomeGroup
0      Aruba      ABW      10.244      78.9      High income
1  Afghanistan      AFG      35.253      5.9      Low income
2      Angola      AGO      45.985      19.1  Upper middle income
```

19) Numerical and Categorical data

```
In [85]: df_numerical_data = df[['BirthRate', 'InternetUsers']]
```

```
In [87]: df_categorical_data = df[['CountryName', 'CountryCode', 'IncomeGroup']]
```

```
In [88]: df_categorical_data.head(7) # it display only category or alphabets words dat
```

```
Out[88]: CountryName CountryCode IncomeGroup
0      Aruba      ABW      High income
1  Afghanistan      AFG      Low income
2      Angola      AGO  Upper middle income
3      Albania      ALB  Upper middle income
4  United Arab Emirates      ARE      High income
5      Argentina      ARG      High income
6      Armenia      ARM  Lower middle income
```

```
In [89]: df_numerical_data.head(7) # It display only numerical data in the table
```

Out[89]:

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985	19.1
3	12.877	57.2
4	11.044	88.0
5	17.716	59.9
6	13.308	41.9

```
In [90]: print(df.shape)
print(df_categorical_data.shape)
print(df_numerical_data.shape)
```

```
(195, 5)
```

```
(195, 3)
```

```
(195, 2)
```

```
In [93]: print(df.columns)
print('=====')
print(df_categorical_data.columns)
print('=====')
print(df_numerical_data.columns)
```

```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')
```

```
=====
```

```
Index(['CountryName', 'CountryCode', 'IncomeGroup'], dtype='object')
```

```
=====
```

```
Index(['BirthRate', 'InternetUsers'], dtype='object')
```

20) index Slicing in pandas

```
In [94]: df[:] # empty slicig it show whole entire data as it is.
```

Out[94]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [95]: `df[0:5] # py know that only this is rows based on index,it show 0 to 4 rows`

Out[95]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [96]: `df[:10]`

Out[96]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
1	Afghanistan	AFG	35.253	5.9000	Low income
2	Angola	AGO	45.985	19.1000	Upper middle income
3	Albania	ALB	12.877	57.2000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0000	High income
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income

In [97]: `df[::-1]` *# it display all rows in reverse*

Out[97]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

In [102... `df[::-5]` *# It displays every -5th rows from bottom = reverse step slicing*

Out[102...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5000	Low income
189	Samoa	WSM	26.172	15.3000	Lower middle income
184	Venezuela, RB	VEN	19.842	54.9000	High income
179	Ukraine	UKR	11.100	41.0000	Lower middle income
174	Trinidad and Tobago	TTO	14.590	63.8000	High income
169	Thailand	THA	11.041	28.9400	Upper middle income
164	Swaziland	SWZ	30.093	24.7000	Lower middle income
159	Sao Tome and Principe	STP	34.537	23.0000	Lower middle income
154	Sierra Leone	SLE	36.729	1.7000	Low income
149	Saudi Arabia	SAU	20.576	60.5000	High income
144	French Polynesia	PYF	16.393	56.8000	High income
139	Papua New Guinea	PNG	28.899	6.5000	Lower middle income
134	Oman	OMN	20.419	66.4500	High income
129	Nicaragua	NIC	20.788	15.5000	Lower middle income
124	Malaysia	MYS	16.805	66.9700	Upper middle income
119	Mongolia	MNG	24.275	20.0000	Upper middle income
114	Macedonia, FYR	MKD	11.222	65.2400	Upper middle income
109	Morocco	MAR	21.023	56.0000	Lower middle income
104	Lesotho	LSO	28.738	5.0000	Lower middle income
99	Liberia	LBR	35.521	3.2000	Low income
94	Kiribati	KIR	29.044	11.5000	Lower middle income
89	Japan	JPN	8.200	89.7100	High income
84	Iceland	ISL	13.400	96.5468	High income
79	Indonesia	IDN	20.297	14.9400	Lower middle income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
74	Hong Kong SAR, China	HKG	7.900	74.2000	High income
69	Grenada	GRD	19.334	35.0000	Upper middle income
64	Guinea	GIN	37.337	1.6000	Low income
59	Micronesia, Fed. Sts.	FSM	23.511	27.8000	Lower middle income
54	Estonia	EST	10.300	79.4000	High income
49	Algeria	DZA	24.738	16.5000	Upper middle income
44	Czech Republic	CZE	10.200	74.1104	High income
39	Cabo Verde	CPV	21.625	37.5000	Lower middle income
34	Cote d'Ivoire	CIV	37.320	8.4000	Lower middle income
29	Central African Republic	CAF	34.076	3.5000	Low income
24	Brazil	BRA	14.931	51.0400	Upper middle income
19	Bosnia and Herzegovina	BIH	9.062	57.7900	Upper middle income
14	Burkina Faso	BFA	40.551	9.1000	Low income
9	Austria	AUT	9.400	80.6188	High income
4	United Arab Emirates	ARE	11.044	88.0000	High income

```
In [98]: df[::20] # get only every 20th row-step slicing
```

Out[98]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
20	Belarus	BLR	12.500	54.1700	Upper middle income
40	Costa Rica	CRI	15.022	45.9600	Upper middle income
60	Gabon	GAB	30.555	9.2000	Upper middle income
80	India	IND	20.291	15.1000	Lower middle income
100	Libya	LBY	21.425	16.5000	Upper middle income
120	Mozambique	MOZ	39.705	5.4000	Low income
140	Poland	POL	9.600	62.8492	High income
160	Suriname	SUR	18.455	37.4000	Upper middle income
180	Uruguay	URY	14.374	57.6900	High income

In [101]:

df[0:100:20]

Out[101]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
20	Belarus	BLR	12.500	54.17	Upper middle income
40	Costa Rica	CRI	15.022	45.96	Upper middle income
60	Gabon	GAB	30.555	9.20	Upper middle income
80	India	IND	20.291	15.10	Lower middle income

In [99]:

df.head(10) # It displays top 10 rows from 0 to 9

Out[99]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9000	High income
1	Afghanistan	AFG	35.253	5.9000	Low income
2	Angola	AGO	45.985	19.1000	Upper middle income
3	Albania	ALB	12.877	57.2000	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0000	High income
5	Argentina	ARG	17.716	59.9000	High income
6	Armenia	ARM	13.308	41.9000	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4000	High income
8	Australia	AUS	13.200	83.0000	High income
9	Austria	AUT	9.400	80.6188	High income

In [100]:

df.head() # It displays top 5 rows from 0 to 4

Out[100...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

21) Check Missing Values --isnull()

- Returns True for missing(none) values
- False for non-missing(value) values

In [113...

```
df.isnull() # It displays if in table contain any null value then it show 'True'
            # It displays if in table contain no any null value then it show 'False'
            # hence as per our table there is no any null value, thus our table become all False
```

Out[113...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [118...

```
df.isna() # We also use short form of null = 'na'
```

Out[118...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

22) Count Missing Values

In [115...

```
df.isnull().sum() # Counts total missing values in each column.
```

Out[115...

```
CountryName      0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
```

23) Remove Missing Values--dropna()

In [119...

```
df.dropna() # It removes rows containing missing (null) values.
            # In our table does not contain any null value hence it can not re
```

Out[119...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

24) Fill Missing Value fillna()

In [120...

```
df.fillna(0) # it replaces missing values with 0
             # In table there is no any missing or null value ,it can't replace
```

Out[120...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

25) Rename Columns rename()

In [122...

```
df.rename(columns={'BirthRate': 'DeathRate'}) # it change the column name into
```

Out[122...

	CountryName	CountryCode	DeathRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [123...

df.head()

Out[123...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [147...

```
df.columns = ['a', 'b', 'c', 'd', 'e']
df.columns
```

Out[147...

Index(['a', 'b', 'c', 'd', 'e'], dtype='object')

In [148...

df.head()

Out[148...

	a	b	c	d	e
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [152...

```
df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeG
df
```

Out[152...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [154...

```
df.columns
```

Out[154...

```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')
```

In [155...

```
df.head()
```

Out[155...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

26) Add new Column and Multiplying two columns

In [224...

df

Out[224...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [140...

```
df['MyCal'] = df.BirthRate * df.InternetUsers
```

In [145...

```
df.head()
```

Out[145...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	Mycalc
0	Aruba	ABW	10.244	78.9	High income	808.2516
1	Afghanistan	AFG	35.253	5.9	Low income	207.9927
2	Angola	AGO	45.985	19.1	Upper middle income	878.3135
3	Albania	ALB	12.877	57.2	Upper middle income	736.5644
4	United Arab Emirates	ARE	11.044	88.0	High income	971.8720

27) Delete Column -- drop()

In [146...

```
df = df.drop('Mycalc', axis=1) # it remove mycalc coulmn by using drop function
df
```

Out[146...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

28) Sorting Data -- sort_values()

In [225...

df

Out[225...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [158...

```
df.sort_values(by = 'InternetUsers') # It sort the values in ascending order.
```

Out[158...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
52	Eritrea	ERI	34.800	0.9000	Low income
172	Timor-Leste	TLS	35.755	1.1000	Lower middle income
11	Burundi	BDI	44.151	1.3000	Low income
156	Somalia	SOM	43.891	1.5000	Low income
117	Myanmar	MMR	18.119	1.6000	Lower middle income
...	...	...	...	...	...
47	Denmark	DNK	10.000	94.6297	High income
163	Sweden	SWE	11.800	94.7836	High income
131	Norway	NOR	11.600	95.0534	High income
22	Bermuda	BMU	10.400	95.3000	High income
84	Iceland	ISL	13.400	96.5468	High income

195 rows × 5 columns

In [159...

```
df.sort_values(by = 'InternetUsers', ascending = False) # it display decending
```

Out[159...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
84	Iceland	ISL	13.400	96.5468	High income
22	Bermuda	BMU	10.400	95.3000	High income
131	Norway	NOR	11.600	95.0534	High income
163	Sweden	SWE	11.800	94.7836	High income
47	Denmark	DNK	10.000	94.6297	High income
...	...	...	...	...	...
64	Guinea	GIN	37.337	1.6000	Low income
156	Somalia	SOM	43.891	1.5000	Low income
11	Burundi	BDI	44.151	1.3000	Low income
172	Timor-Leste	TLS	35.755	1.1000	Lower middle income
52	Eritrea	ERI	34.800	0.9000	Low income

195 rows × 5 columns

29) Filtering Data

In [161...

```
df[df['BirthRate']>11] # Shows birthrate value greater than 11.
```

Out[161...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

159 rows × 5 columns

In [217...

```
len(df[df['BirthRate']>11])
```

Out[217...

159

In [162...

```
df[df['InternetUsers']<50] # Shows internetusers value less than 50.
```

Out[162...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
6	Armenia	ARM	13.308	41.9	Lower middle income
11	Burundi	BDI	44.151	1.3	Low income
13	Benin	BEN	36.440	4.9	Low income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

115 rows × 5 columns

In [219...

```
len(df[df['InternetUsers'] < 50])
```

Out[219...

115

30) Unique Values--unique()

In [163...

```
df['CountryName'].unique() # Returns unique names.
```

```
Out[163... array(['Aruba', 'Afghanistan', 'Angola', 'Albania',
      'United Arab Emirates', 'Argentina', 'Armenia',
      'Antigua and Barbuda', 'Australia', 'Austria', 'Azerbaijan',
      'Burundi', 'Belgium', 'Benin', 'Burkina Faso', 'Bangladesh',
      'Bulgaria', 'Bahrain', 'Bahamas, The', 'Bosnia and Herzegovina',
      'Belarus', 'Belize', 'Bermuda', 'Bolivia', 'Brazil', 'Barbados',
      'Brunei Darussalam', 'Bhutan', 'Botswana',
      'Central African Republic', 'Canada', 'Switzerland', 'Chile',
      'China', 'Cote d'Ivoire', 'Cameroon', 'Congo, Rep.', 'Colombia',
      'Comoros', 'Cabo Verde', 'Costa Rica', 'Cuba', 'Cayman Islands',
      'Cyprus', 'Czech Republic', 'Germany', 'Djibouti', 'Denmark',
      'Dominican Republic', 'Algeria', 'Ecuador', 'Egypt, Arab Rep.',
      'Eritrea', 'Spain', 'Estonia', 'Ethiopia', 'Finland', 'Fiji',
      'France', 'Micronesia, Fed. Sts.', 'Gabon', 'United Kingdom',
      'Georgia', 'Ghana', 'Guinea', 'Gambia, The', 'Guinea-Bissau',
      'Equatorial Guinea', 'Greece', 'Grenada', 'Greenland', 'Guatemala',
      'Guam', 'Guyana', 'Hong Kong SAR, China', 'Honduras', 'Croatia',
      'Haiti', 'Hungary', 'Indonesia', 'India', 'Ireland',
      'Iran, Islamic Rep.', 'Iraq', 'Iceland', 'Israel', 'Italy',
      'Jamaica', 'Jordan', 'Japan', 'Kazakhstan', 'Kenya',
      'Kyrgyz Republic', 'Cambodia', 'Kiribati', 'Korea, Rep.', 'Kuwait',
      'Lao PDR', 'Lebanon', 'Liberia', 'Libya', 'St. Lucia',
      'Liechtenstein', 'Sri Lanka', 'Lesotho', 'Lithuania', 'Luxembourg',
      'Latvia', 'Macao SAR, China', 'Morocco', 'Moldova', 'Madagascar',
      'Maldives', 'Mexico', 'Macedonia, FYR', 'Mali', 'Malta', 'Myanmar',
      'Montenegro', 'Mongolia', 'Mozambique', 'Mauritania', 'Mauritius',
      'Malawi', 'Malaysia', 'Namibia', 'New Caledonia', 'Niger',
      'Nigeria', 'Nicaragua', 'Netherlands', 'Norway', 'Nepal',
      'New Zealand', 'Oman', 'Pakistan', 'Panama', 'Peru', 'Philippines',
      'Papua New Guinea', 'Poland', 'Puerto Rico', 'Portugal',
      'Paraguay', 'French Polynesia', 'Qatar', 'Romania',
      'Russian Federation', 'Rwanda', 'Saudi Arabia', 'Sudan', 'Senegal',
      'Singapore', 'Solomon Islands', 'Sierra Leone', 'El Salvador',
      'Somalia', 'Serbia', 'South Sudan', 'Sao Tome and Principe',
      'Suriname', 'Slovak Republic', 'Slovenia', 'Sweden', 'Swaziland',
      'Seychelles', 'Syrian Arab Republic', 'Chad', 'Togo', 'Thailand',
      'Tajikistan', 'Turkmenistan', 'Timor-Leste', 'Tonga',
      'Trinidad and Tobago', 'Tunisia', 'Turkey', 'Tanzania', 'Uganda',
      'Ukraine', 'Uruguay', 'United States', 'Uzbekistan',
      'St. Vincent and the Grenadines', 'Venezuela, RB',
      'Virgin Islands (U.S.)', 'Vietnam', 'Vanuatu',
      'West Bank and Gaza', 'Samoa', 'Yemen, Rep.', 'South Africa',
      'Congo, Dem. Rep.', 'Zambia', 'Zimbabwe'], dtype=object)
```

```
In [164... df['BirthRate'].unique() # Returns unique values of birthrate.
```

```
Out[164...] array([10.244, 35.253, 45.985, 12.877, 11.044, 17.716, 13.308, 16.447,
      13.2 , 9.4 , 18.3 , 44.151, 11.2 , 36.44 , 40.551, 20.142,
      9.2 , 15.04 , 15.339, 9.062, 12.5 , 23.092, 10.4 , 24.236,
      14.931, 12.188, 16.405, 18.134, 25.267, 34.076, 10.9 , 10.2 ,
      13.385, 12.1 , 37.32 , 37.236, 37.011, 16.076, 34.326, 21.625,
      15.022, 11.436, 8.5 , 25.486, 10. , 21.198, 24.738, 21.07 ,
      28.032, 34.8 , 9.1 , 10.3 , 32.925, 10.7 , 20.463, 12.3 ,
      23.511, 30.555, 12.2 , 13.332, 33.131, 37.337, 42.525, 37.503,
      35.362, 19.334, 14.5 , 27.465, 17.389, 18.885, 7.9 , 21.593,
      25.345, 20.297, 20.291, 15. , 17.9 , 31.093, 13.4 , 21.3 ,
      13.54 , 27.046, 8.2 , 22.73 , 35.194, 27.2 , 24.462, 29.044,
      8.6 , 20.575, 27.051, 13.426, 35.521, 21.425, 15.43 , 17.863,
      28.738, 10.1 , 11.3 , 11.256, 21.023, 12.141, 34.686, 21.447,
      19.104, 11.222, 44.138, 9.5 , 18.119, 11.616, 24.275, 39.705,
      33.801, 39.459, 16.805, 29.937, 17. , 49.661, 40.045, 20.788,
      11.6 , 20.923, 13.12 , 20.419, 29.582, 19.68 , 20.198, 23.79 ,
      28.899, 9.6 , 10.8 , 21.588, 16.393, 11.94 , 8.8 , 32.689,
      20.576, 33.477, 38.533, 9.3 , 30.578, 36.729, 17.476, 43.891,
      37.126, 34.537, 18.455, 11.8 , 30.093, 18.6 , 24.043, 45.745,
      36.08 , 11.041, 30.792, 21.322, 35.755, 25.409, 14.59 , 19.8 ,
      16.836, 39.518, 43.474, 11.1 , 14.374, 22.5 , 16.306, 19.842,
      15.537, 26.739, 30.394, 26.172, 32.947, 20.85 , 42.394, 40.471,
      35.715])
```

31) Count Unique Values--nunique()

```
In [226...] df
```

```
Out[226...] df
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [165... df['CountryName'].nunique() # counts unique values,avoid non unique(nunique)
```

```
Out[165... 195
```

```
In [166... df['CountryCode'].nunique() # in this table no any non unique values is present,
```

```
Out[166... 195
```

32) GroupBy Function

```
In [167... df.head()
```

```
Out[167... 
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

```
In [174... df.groupby('CountryCode')['BirthRate'].mean() # two columns arranged together i
```

```
Out[174... CountryCode
ABW    10.244
AFG    35.253
AGO    45.985
ALB    12.877
ARE    11.044
...
WSM    26.172
YEM    32.947
ZAF    20.850
ZMB    40.471
ZWE    35.715
Name: BirthRate, Length: 195, dtype: float64
```

33) Apply function--apply()

```
In [227... df
```

Out[227...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [176...

```
df['InternetUsers'].apply(lambda x: x + 5) # It just apply addition '5' in each
```

Out[176...

```
0      83.9
1      10.9
2      24.1
3      62.2
4      93.0
...
190    25.0
191    51.5
192     7.2
193    20.4
194    23.5
Name: InternetUsers, Length: 195, dtype: float64
```

In [177...

```
df['InternetUsers'].apply(lambda x: x - 5) # # It just apply subtraction '-5' in each
```

Out[177...

```
0      73.9
1       0.9
2      14.1
3      52.2
4      83.0
...
190    15.0
191    41.5
192    -2.8
193    10.4
194    13.5
Name: InternetUsers, Length: 195, dtype: float64
```



```
In [180...] df['InternetUsers'].apply(lambda x: x * 5)  ## It just apply multiply '5' in ea
```

```
Out[180...] 0      394.5
            1       29.5
            2      95.5
            3     286.0
            4     440.0
            ...
           190    100.0
           191    232.5
           192     11.0
           193     77.0
           194     92.5
            Name: InternetUsers, Length: 195, dtype: float64
```

```
In [181...] df['InternetUsers'].apply(lambda x: x / 5)  ## It just apply divide '5' in each
```

```
Out[181...] 0      15.78
            1       1.18
            2       3.82
            3      11.44
            4      17.60
            ...
           190       4.00
           191       9.30
           192       0.44
           193       3.08
           194       3.70
            Name: InternetUsers, Length: 195, dtype: float64
```

34) Merge Data Frames--merge()

```
In [182...] df
```

Out[182...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

here we apply filter to two value at a same time

In [221...

```
df[(df.BirthRate > 30) & (df.InternetUsers < 2)] # in this methode we combine b
```

Out[221...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
52	Eritrea	ERI	34.800	0.9	Low income
55	Ethiopia	ETH	32.925	1.9	Low income
64	Guinea	GIN	37.337	1.6	Low income
127	Niger	NER	49.661	1.7	Low income
154	Sierra Leone	SLE	36.729	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income
172	Timor-Leste	TLS	35.755	1.1	Lower middle income

In [222...

df

Out[222...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [211...

```
# Combines two DataFrames values using common column.

import pandas as pd

pd.DataFrame()
# first Dataframe
Country= pd.DataFrame({
    'InternetUsers' : [78.9,5.9,19.1,57.2,88.0],
    'CountryName': ['Aruba','Afghanistan','Angola','Albania','United Arab Emirats']
})
# Second dataframe
Rate= pd.DataFrame({
    'InternetUsers' : [78.9,5.9,19.1,57.2,88.0],
    'BirthRate': [10.244,35.253,45.985,12.877,11.044]
})
# merge dataframe
result = pd.merge(Country, Rate, on = 'InternetUsers')

print(result) # Here it Combines two DataFrames using common column.
```

	InternetUsers	CountryName	BirthRate
0	78.9	Aruba	10.244
1	5.9	Afghanistan	35.253
2	19.1	Angola	45.985
3	57.2	Albania	12.877
4	88.0	United Arab Emirates	11.044

35) Concatenate DataFrames – concat()

In [202...

df

Out[202...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [212...

```
pd.concat([Country,Rate]) # Joins DataFrames vertically.
```

Out[212...

	InternetUsers	CountryName	BirthRate
0	78.9	Aruba	NaN
1	5.9	Afghanistan	NaN
2	19.1	Angola	NaN
3	57.2	Albania	NaN
4	88.0	United Arab Emirates	NaN
0	78.9	NaN	10.244
1	5.9	NaN	35.253
2	19.1	NaN	45.985
3	57.2	NaN	12.877
4	88.0	NaN	11.044

36) Remove Duplicate Rows – drop_duplicates()

In [214...

df

Out[214...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [213...

```
df.drop_duplicates() # It Removes all repeated rows.
```

Out[213...

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In []:

37) Basic Pandas Functions Summary Table

All Pandas Function and their Purpose

read_csv() Read CSV file

head() Show first rows

tail() Show last rows

shape Total rows and columns

columns Show column names

dtypes Show data types

info() Dataset information

describe() Statistical summary

isnull() Detect missing values

dropna() Remove missing values

fillna() Fill missing values

sort_values() Sort data

value_counts() Count values

groupby() Group data

merge() Combine DataFrames

concat() Join DataFrames

drop_duplicates() Remove duplicates

to_csv() Save CSV

=====

Task : Pandas Basic coding Complete

=====

In []:

In []:

In []: